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$$f(x) = \frac{\sin x - \cos x}{\sin x + \cos x}$$
$$f'(x) = \frac{(\cos x + \sin x)(\sin x + \cos x) - (\sin x - \cos x)(\cos x - \sin x)}{(\sin x + \cos x)^2}$$
$$= \frac{(\cos x + \sin x)^2 + (\sin x - \cos x)^2}{(\sin x + \cos x)^2}$$
$$= \frac{\cos^2 x + 2\cos x \sin x + \sin^2 x + \sin^2 x - 2\sin x \cos x + \cos^2 x}{\sin^2 x + 2\cos x \sin x + \cos^2 x}$$
$$= \frac{2}{1 + \sin(2x)}$$

References